Pricing Credit Derivatives with Counterparty Risk

Leila Rahimi

Allameh Tabataba'i University, 2016

A summary of my Master's thesis in Financial Mathematics

About this thesis

This was my Master's thesis in Financial Mathematics at Allameh Tabataba'i University, supervised by Prof. Mohammad Jelodari Mamaghani and advised by Dr. Shekufeh Bani Hashemi. The topic is how to price a derivative contract when there is a real chance that *either side* of the deal might not pay what they owe.

Most option pricing theory assumes the two parties in a contract will always honor their side of the deal. That assumption fell apart in a very visible way during the 2008 financial crisis, when it turned out that the counterparty on the other end of a swap or a credit derivative could itself go under. This thesis works through how to price a claim once you stop assuming that away, for both sides of the contract.

The problem in plain terms

Say two parties, A and B, enter into a derivative contract. Normally we price this contract assuming both A and B will pay whatever they owe at maturity. But what if B might go bankrupt before then? Or what if A might? The value of the contract to each side now depends on the chance that the other one defaults, and on how much money would actually be recovered if that happened.

This is called **counterparty risk**: the risk that the entity on the other side of *your own trade* fails to pay, as opposed to ordinary credit risk, where the risk sits in a bond or a loan you're holding. Counterparty risk shows up in over-the-counter contracts like swaps and credit default swaps, where there's no exchange standing in the middle to guarantee payment.

The goal of the thesis is to find a pricing formula for a claim between two parties, where either one could default at any point before the contract matures, and to do it from the point of view of both A and B.

Background: two families of credit risk models

Before getting to the main problem, the thesis reviews how credit risk has traditionally been modeled. There are two main approaches.

Structural models. These start with Merton's 1974 model, which builds directly on Black-Scholes-Merton option pricing. The idea is simple: a firm defaults when the value of its assets drops below its liabilities. Default is treated like a put option being exercised. Later researchers (Black and Cox, Longstaff and Schwartz, Leland and Toft, and others) extended this by allowing default to happen at any time before maturity, not just at the debt's maturity date, and by adding things like stochastic interest rates and taxes. The well-known commercial KMV/Moody's model also comes from this family.

Structural models are intuitive, but they have real weaknesses: you need to estimate the market value of a firm's assets, which usually isn't directly observable, and the model treats default as something that approaches gradually and predictably, which doesn't match how default actually looks in the data.

Reduced-form (intensity) models. These came later, from Jarrow and Turnbull (1995) and Duffie and Singleton (1999) among others. Instead of modeling *why* a firm defaults, these models treat default as a sudden, unpredictable jump — like a Poisson process — governed by a "hazard rate" or default intensity, usually written as λ. You don't need to know anything about the firm's asset value or capital structure; you just need a process for how likely default is at each moment, calibrated from market prices (typically bond spreads).

This thesis builds on the reduced-form approach, mainly following the framework of Duffie and Huang (1996) and Duffie and Singleton (1999).

Two simple building blocks

The thesis walks through both of the classic reduced-form models before building the bilateral counterparty-risk model on top of them:

- **Jarrow-Turnbull model.** Default and recovery are modeled with a Poisson process. The simplest version assumes recovery is only paid at maturity, not at the moment of default. A nice feature of this model is that it's very easy to calibrate to observed bond prices — the thesis works through a numerical example, backing out implied default probabilities for two zero-coupon-like bonds.
- **Duffie-Singleton model.** This is more realistic: recovery can happen at any time, not just at maturity, and the recovery amount is modeled as a fraction of what the bond would have been worth had there been no default (this is called "recovery of market value," or RMV). The thesis also works through a numerical example here, pricing a two-year coupon bond on a binomial tree under random interest rates and a 4% annual default probability.

The core contribution: pricing under bilateral counterparty risk

With that groundwork in place, the thesis turns to the actual question: what is the contract worth, from each side's perspective, once *both* parties can default?

The setup: A buys a claim from B with payoff $\Pi(X_T)$ at maturity T, assuming neither side has defaulted before then. Each party has its own default intensity and its own recovery rate. The key modeling choice is what happens to the value of the contract the instant one side defaults:

- If A is owed money (the contract's value to A is positive) and B defaults, A only recovers a fraction of what's owed — A takes the loss.
- If A owes money (the contract's value to A is negative) and B defaults, nothing changes for A — A still has to pay B in full, debt doesn't disappear just because the other side defaulted.
- The same logic applies symmetrically if A is the one who defaults.

In other words, whoever is "in the money" when the other side defaults takes a haircut; whoever is "out of the money" doesn't get let off the hook.

Working this through (using a martingale argument and the Duffie-Huang recursive valuation method), the thesis arrives at a pricing formula where the discount rate used isn't just the risk-free rate — it's the risk-free rate *plus* an extra spread that switches depending on the sign of the contract's own value:

> *discount rate = risk-free rate + (A's default spread, if the contract is worth less than zero) + (B's default spread, if the contract is worth zero or more)*

This is the central result of the thesis, derived first in discrete time and then extended to continuous time using the Feynman-Kac connection between expectations and PDEs.

Why this is harder than it looks

Here's the catch: the discount rate in that formula depends on the *sign of the value you're trying to compute*. The pricing equation isn't the usual linear Black-Scholes-type PDE; it's a non-linear PDE, because the equation for V depends on V itself through this indicator-function switch. That non-linearity means the price at any point in time technically depends on the whole path the value process has taken, not just on where things stand right now. This makes the formula very awkward to use in practice — it's not the kind of clean, closed-form expression that's easy to compute or hedge with.

Removing the path dependence

The last technical chapter of the thesis tackles that problem directly: how do you get rid of the path dependence so the formula becomes usable?

The approach is a change of measure. Under some technical restrictions on how the short-term interest rate is allowed to behave, the thesis constructs a new probability measure (via a Radon-Nikodym derivative) under which a transformed value function becomes a proper martingale. The trick is choosing a function f, applied to the value process, such that f of the value process is a martingale under the new measure — and once that's true, the non-linear part of the problem essentially disappears and you're left with a closed-form, non-path-dependent expression.

The thesis works out the explicit form of this transformation (an ODE that f has to solve) and then applies it to a worked example: a one-sided case where only counterparty B can default, with a default intensity that's a simple power function of the current contract value. In that special case, the change-of-measure trick produces a clean closed-form price.

Why this matters

Counterparty risk pricing isn't an academic curiosity — it's directly tied to how derivatives desks price CVA/DVA (credit and debit valuation adjustments), and to how OTC markets responded to 2008 by pushing for collateral agreements and central clearing. Knowing how to price a claim when *both* sides of the trade carry default risk, and having a tractable (non-path-dependent) way to do it, is exactly the kind of problem that sits behind those adjustments.

What I'd do next

The thesis closes by pointing at a few open directions:

- Implementing the pricing PDEs numerically (the thesis mentions finite-difference approaches used elsewhere in the literature, like Crank-Nicolson) for cases that don't reduce to closed form.

- Extending the bilateral model to more than two counterparties, or to portfolios of correlated claims, which is closer to how real default risk shows up in practice.
- Testing how sensitive the price is to the choice of recovery assumption (recovery of market value versus other recovery conventions), since the thesis shows these aren't interchangeable.

References

The thesis leans heavily on:

- Duffie, D. and Huang, M. (1996), *Swap rates and credit quality*, Journal of Finance.
- Duffie, D., Pan, J. and Singleton, K. (2000), *Transform analysis and asset pricing for affine jump-diffusions*, Econometrica.
- Duffie, D. and Singleton, K. (1999/2003), *Credit Risk*, Princeton University Press.
- Jarrow, R. and Turnbull, S. (1995), *Pricing derivatives on financial securities subject to default risk*, Journal of Finance.
- Merton, R. (1974), *On the pricing of corporate debt*, Journal of Finance.
- Kim, J. and Leung, T. (2015), *Pricing derivatives with counterparty risk and collateralization*, European Journal of Operational Research.

A full reference list is in the original thesis.

Original thesis (in Persian): "Pricing Credit Derivatives with Counterparty Risk," Allameh Tabataba'i University, Faculty of Mathematical Sciences, Winter 2016. Supervisor: Mohammad Jelodari Mamaghani. Advisor: Shekufeh Bani Hashemi.